UCLA Math 151A (Fall 2025) — Homework 2

Problem 1

Let

$$f(x) = \cos x + \sin^2(50x), \qquad p = \frac{\pi}{2}.$$

Then

$$f'(x) = -\sin x + 50\sin(100x), \qquad f''(x) = -\cos x + 5000\cos(100x).$$

In particular,

$$f'(p) = -1, \qquad f''(p) = 5000.$$

1. The local Newton theorem and why $1/M$ appears

Write the error as $e_k := x_k - p$. A standard local result (proved via Taylor with remainder and the mean value theorem) is:

$$|e_{k+1}| \;\le\; \frac{\max_{x\in I}|f''(x)|}{2\,\min_{x\in I}|f'(x)|}\,|e_k|^2 \;=\; M\,|e_k|^2 \quad \text{for all } x_k \in I,$$

provided $f \in C^2(I)$ and $\min_{x\in I}|f'(x)| > 0$. If the initial error satisfies $M|e_0| < 1$ and (crucially) the iterates remain in I, then

$$|e_1| \le M|e_0|^2 \le |e_0|, \quad |e_2| \le M|e_1|^2 \le M|e_0|\,|e_1| \le |e_1|, \;\ldots$$

so the errors decrease monotonically to 0, i.e., Newton converges to p and is locally quadratic. Hence the practical sufficient condition is

$$|x_0 - p| \;<\; \frac{1}{M}, \qquad M := \frac{\max_{x\in I}|f''(x)|}{2\min_{x\in I}|f'(x)|}.$$

This explains the appearance of $1/M$ in the hint: it is the radius of the local quadratic basin inherited from the inequality above.

2. Choosing a small interval I and bounding f', f''

Write $x = p + h$ with $|h|$ small. Using

$$\cos(\tfrac{\pi}{2}+h) = -\sin h, \quad \sin\big(100(\tfrac{\pi}{2}+h)\big) = \sin(100h), \quad \cos\big(100(\tfrac{\pi}{2}+h)\big) = \cos(100h),$$

we have

$$f'(p+h) = -\cos h + 50\sin(100h), \qquad f''(p+h) = \sin h + 5000\cos(100h).$$

For $|h| \le \delta$, the elementary inequalities

$$\cos h \ge 1 - \frac{h^2}{2}, \qquad |\sin(100h)| \le 100|h|, \qquad |\sin h| \le |h|, \qquad |\cos(100h)| \le 1$$

give the *usable* bounds

$$\boxed{|f'(p+h)| \;\ge\; \big|\,|\cos h| - 50|\sin(100h)|\,\big| \;\ge\; 1 - \frac{h^2}{2} - 5000|h|},$$

$$\boxed{|f''(p+h)| \;\le\; |\sin h| + 5000|\cos(100h)| \;\le\; \delta + 5000}.$$

Numerical choice. Take $I = [p-\delta, p+\delta]$ with $\delta = 10^{-4}$. Then

$$\min_I |f'(x)| \;\gtrsim\; 1 - 5000\delta - \tfrac{\delta^2}{2} \approx 1 - 0.5 - 5 \times 10^{-9} \approx 0.5,$$

$$\max_I |f''(x)| \;\le\; 5000 + \delta \approx 5000.0001.$$

Hence

$$M \;\lesssim\; \frac{5000.0001}{2 \cdot 0.5} \approx 5000.0001, \qquad \frac{1}{M} \;\gtrsim\; 2.0 \times 10^{-4}.$$

Refinement (grid computation). On a dense grid of 40001 points over I, we found

$$\min_I |f'(x)| \approx 0.50000833, \qquad \max_I |f''(x)| \approx 5000.00000001,$$

so

$$M \approx 4999.9167, \qquad \frac{1}{M} \approx 2.0000333 \times 10^{-4}.$$

Thus, a rough but sharp practical threshold is

$$\boxed{|x_0 - p| \;\lesssim\; 2.0 \times 10^{-4}}.$$

3. Self-consistency of the interval I

A subtle but important point: to *guarantee* iterates remain in I, one typically requires the entire ball of radius $1/M$ around p to be contained in I. With the symmetric choice $\delta = 10^{-4}$ we have $1/M \approx 2.0 \times 10^{-4} > \delta$. Therefore a fully self-contained guarantee using this I is to restrict the initial error by

$$|x_0 - p| \;\le\; \min\{\delta,\, 1/M\} = \delta = 10^{-4}.$$

This is only slightly more conservative than $1/M$ and leaves the qualitative conclusion unchanged. (Choosing a wider *symmetric* interval is impossible here without crossing a zero of f' near $p + 2 \times 10^{-4}$; see below.)

4. Asymmetry and why the right side is fragile

Using $f'(p+h) = -\cos h + 50\sin(100h) \approx -\big(1 - \frac{h^2}{2}\big) + 5000h$, the linear term predicts a sign change of f' near $h \approx 1/5000 = 2 \times 10^{-4}$. This explains why starting slightly to the *right* of p can fail unless $|x_0 - p|$ is below roughly 2×10^{-4}, while the *left* side is more forgiving.

5. Empirical check (Newton from $x_0 = p + h$)

We ran Newton with tolerance 10^{-12} and tested a range of offsets h. A representative subset:

$h = x_0 - p$	converged to p?	iters	final x
$-3.0 \cdot 10^{-4}$	Yes	6	1.570796
$-2.2 \cdot 10^{-4}$	Yes	6	1.570796
$-2.05 \cdot 10^{-4}$	Yes	6	1.570796
$-2.0 \cdot 10^{-4}$	Yes	6	1.570796
$-1.8 \cdot 10^{-4}$	Yes	6	1.570796
$-1.0 \cdot 10^{-4}$	Yes	5	1.570796
$+1.0 \cdot 10^{-4}$	Yes	6	1.570796
$+1.8 \cdot 10^{-4}$	Yes	8	1.570796
$+1.9 \cdot 10^{-4}$	Yes	9	1.570796
$\boxed{+2.0 \cdot 10^{-4}}$	**No**	50	0.139998
$+2.05 \cdot 10^{-4}$	**No**	10	1.571196
$+2.2 \cdot 10^{-4}$	**No**	8	1.571196
$+2.5 \cdot 10^{-4}$	**No**	7	1.571196
$+3.0 \cdot 10^{-4}$	**No**	6	1.571196

This matches the theory: the curvature-based threshold $1/M \approx 2 \times 10^{-4}$ is tight on the right, while the left side tolerates larger offsets.

```
import numpy as np
from math import sin, cos, pi

p = np.pi/2
f  = lambda x: np.cos(x) + np.sin(50*x)**2
fp = lambda x: -np.sin(x) + 50*np.sin(100*x)
fpp= lambda x: -np.cos(x) + 5000*np.cos(100*x)

# Bounds on I = [p-delta, p+delta]
delta = 1e-4
xs = np.linspace(p-delta, p+delta, 40001)
min_fp = np.min(np.abs(fp(xs)))
max_fpp = np.max(np.abs(fpp(xs)))
M = max_fpp/(2*min_fp)
print("min|f'|   ",min_fp,"␣max|f''|   ",max_fpp,"␣1/ M  ",1/M)

# Newton iteration from x0 = p+h
def newton(x0, maxit=50, tol=1e-12):
    x = x0
    for k in range(maxit):
        d = fp(x)
        if d == 0: break
        xnew = x - f(x)/d
        if abs(xnew-x) < tol: return xnew, k+1, True
        x = xnew
    return x, k+1, False
```

Problem 2

1. Algebraic equivalence

Starting with the standard secant update

$$p_n \;=\; p_{n-1} - f(p_{n-1})\,\frac{p_{n-1} - p_{n-2}}{f(p_{n-1}) - f(p_{n-2})},$$

expand the numerator and collect terms over the common denominator to get the *barycentric* form

$$\boxed{\; p_n \;=\; \frac{f(p_{n-1})\,p_{n-2} - f(p_{n-2})\,p_{n-1}}{f(p_{n-1}) - f(p_{n-2})} \;}.$$

This is an exact algebraic identity.

2. Floating-point cancellation: which is safer when?

Let $\Delta p := p_{n-1} - p_{n-2}$, $\Delta f := f(p_{n-1}) - f(p_{n-2})$. Under the standard rounding model $\mathrm{fl}(a \circ b) = (a \circ b)(1 + \delta)$, $|\delta| \lesssim \varepsilon$:

- **Case A:** $p_{n-1} \approx p_{n-2}$ ($|\Delta p|$ tiny) while $|\Delta f|$ is not tiny.
  The *barycentric* form avoids subtracting nearly equal p's; its numerator uses two products fp of normal size, then one subtraction. The standard form explicitly forms Δp, which may be dominated by rounding.
  Safer: barycentric form.
- **Case B:** $f(p_{n-1}) \approx f(p_{n-2})$ ($|\Delta f|$ tiny).
  In the barycentric form the numerator $f_{n-1}p_{n-2} - f_{n-2}p_{n-1}$ subtracts two quantities of size $\sim p \cdot f$, causing severe cancellation when $f_{n-1} \approx f_{n-2}$. The standard form constructs the ratio $\Delta p/\Delta f$ (large), but the outer subtraction $p_{n-1} - f_{n-1}(\Delta p/\Delta f)$ is not a difference of nearly equal large numbers.
  Safer: standard form.

3. A simple hybrid rule (double precision)

With $\varepsilon \approx 2.22 \times 10^{-16}$, define

$$\tau_p = \sqrt{\varepsilon}\,\max\{1, |p_{n-1}|, |p_{n-2}|\}, \qquad \tau_f = \sqrt{\varepsilon}\,\max\{|f_{n-1}|, |f_{n-2}|\}.$$

- If $|\Delta f| \le \tau_f$: use the *standard* form $p_n = p_{n-1} - f_{n-1}\,\Delta p/\Delta f$.
- Else if $|\Delta p| \le \tau_p$: use the *barycentric* form $p_n = \dfrac{f_{n-1}p_{n-2} - f_{n-2}p_{n-1}}{\Delta f}$.
- Else: default to the standard form.

Heuristically, $\sqrt{\varepsilon}$ balances rounding vs. truncation and is a standard scale for switching tolerances.

Problem 3

For $f(x) = x^2 - 1$ and $f'(x) = 2x$, Newton gives

$$x_{n+1} = x_n - \frac{x_n^2 - 1}{2x_n} = \frac{x_n^2 + 1}{2x_n} = \frac{1}{2}\Big(x_n + \frac{1}{x_n}\Big).$$

With $x_0 = 100{,}000$, the first 10 iterates and error ratios are:

n	x_n	$\lvert e_n\rvert = \lvert x_n - 1\rvert$	$r_n = \dfrac{\lvert e_{n+1}\rvert}{\lvert e_n\rvert}$	$q_n = \dfrac{\lvert e_{n+1}\rvert}{\lvert e_n\rvert^2}$
0	100000.000000	99999.000000	0.499995	5.0×10^{-6}
1	50000.000005	49999.000005	0.499990	1.0×10^{-5}
2	25000.000013	24999.000013	0.499980	2.0×10^{-5}
3	12500.000026	12499.000026	0.499960	4.0×10^{-5}
4	6250.000053	6249.000053	0.499920	8.0×10^{-5}
5	3125.000107	3124.000107	0.499840	1.6×10^{-4}
6	1562.500213	1561.500213	0.499680	3.2×10^{-4}
7	781.250427	780.250427	0.499360	6.4×10^{-4}
8	390.625853	389.625853	0.498720	1.28×10^{-3}
9	195.314207	194.314207	0.497440	2.56×10^{-3}

Why the two phases occur: exact error recursion

Let $e_n := x_n - 1$. Then

$$x_{n+1} - 1 = \frac{x_n^2 + 1}{2x_n} - 1 = \frac{(x_n - 1)^2}{2x_n} = \frac{e_n^2}{2(1 + e_n)},$$

so

$$\boxed{e_{n+1} = \frac{e_n^2}{2(1 + e_n)}}$$

$$\Rightarrow \quad r_n = \frac{|e_{n+1}|}{|e_n|} = \frac{|e_n|}{2|1 + e_n|}, \qquad q_n = \frac{|e_{n+1}|}{|e_n|^2} = \frac{1}{2|1 + e_n|}.$$

- If $|e_n| \gg 1$: $r_n \approx \frac{|e_n|}{2|e_n|} \approx \frac{1}{2}$ (apparent *linear* halving), and $q_n \approx \frac{1}{2|e_n|}$ tiny (matches the table: $q_0 \approx 5 \cdot 10^{-6}$).
- If $|e_n| \ll 1$: $r_n \to 0$ and $q_n \to \frac{1}{2}$, i.e., *quadratic* convergence with asymptotic constant 1/2.

This agrees with the general Newton theory at a simple root p: if $f'(p) \neq 0$, then

$$e_{n+1} = \frac{f''(\xi_n)}{2f'(p)}\, e_n^2 \quad \text{for some } \xi_n \to p,$$

so the order is 2 and the asymptotic constant is $f''(p)/(2f'(p))$. Here $f'(1) = 2$, $f''(1) = 2$, hence $f''(1)/(2f'(1)) = 1/2$.

```
# Newton for f(x)=x^2-1, x0=100000
x = 100000.0
xs = [x]
for _ in range(40):
    x = 0.5*(x + 1.0/x)
    xs.append(x)

# Derived exact recursion: e_{n+1} = e_n^2 / (2*(1+e_n))
```